

1.Read and display array elements

```
data structure.cpp  x
1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i;
6
7      printf("Enter number of elements: ");
8      scanf("%d", &n);
9
10     printf("Enter array elements:\n");
11     for(i = 0; i < n; i++)
12     {
13         scanf("%d", &a[i]);
14     }
15
16     printf("Array elements are:\n");
17     for(i = 0; i < n; i++)
18     {
19         printf("%d ", a[i]);
20     }
21
22     return 0;
23 }
```

```
C:\Users\SRIRAM\OneDrive\D  x  +  v
Enter number of elements: 5
Enter array elements:
12 32 43 56 87
Array elements are:
12 32 43 56 87
-----
Process exited after 21.65 seconds with return value 0
Press any key to continue . . . |
```

2.Find sum and average of array elements

```
data structure.cpp  ^
1  #include <stdio.h>
2
3  int main()
4  {
5      int a[100], n, i, sum = 0;
6      float average;
7
8      printf("Enter number of elements: ");
9      scanf("%d", &n);
10
11     printf("Enter array elements:\n");
12     for(i = 0; i < n; i++)
13     {
14         scanf("%d", &a[i]);
15         sum = sum + a[i];
16     }
17
18     average = (float)sum / n;
19
20     printf("Sum = %d\n", sum);
21     printf("Average = %.2f\n", average);
22
23     return 0;
24 }
```

```
C:\Users\SRIRAM\OneDrive\D  x  +  v
Enter number of elements: 5
Enter array elements:
12 34 56 78 43
Sum = 223
Average = 44.60
-----
Process exited after 77.21 seconds with return value 0
Press any key to continue . . . |
```

3. Find largest and smallest element in an array

```
ata structure.cpp
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, i, largest, smallest;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter array elements:\n");
11    for(i = 0; i < n; i++)
12    {
13        scanf("%d", &a[i]);
14    }
15
16    largest = a[0];
17    smallest = a[0];
18
19    for(i = 1; i < n; i++)
20    {
21        if(a[i] > largest)
22            largest = a[i];
23
24        if(a[i] < smallest)
25            smallest = a[i];
26    }
27
28    printf("Largest element = %d\n", largest);
29    printf("Smallest element = %d\n", smallest);
30
31    return 0;
32 }
```

```
C:\Users\SRIRAM\OneDrive\D x + v
Enter number of elements: 5
Enter array elements:
21 32 45 56 87
Largest element = 87
Smallest element = 21

-----
Process exited after 10.97 seconds with return value 0
Press any key to continue . . . |
```

4. Count even and odd elements in an array

```
#include <stdio.h>

int main()
{
    int a[100], n, i;
    int even = 0, odd = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);

        if(a[i] % 2 == 0)
            even++;
        else
            odd++;
    }

    printf("Even elements count = %d\n", even);
    printf("Odd elements count = %d\n", odd);

    return 0;
}
```

```
C:\Users\SRIRAM\OneDrive\D x + v
Enter number of elements: 5
Enter array elements:
12 23 45 67 78
Even elements count = 2
Odd elements count = 3

-----
Process exited after 10.59 seconds with return value 0
Press any key to continue . . . |
```

5.Reverse an array

```
#include <stdio.h>
```

```
int main()
```

```
{  
    int a[100], n, i;
```

```
    printf("Enter number of elements: ");  
    scanf("%d", &n);
```

```
    printf("Enter array elements:\n");  
    for(i = 0; i < n; i++)  
    {  
        scanf("%d", &a[i]);  
    }
```

```
    printf("Reversed array:\n");  
    for(i = n - 1; i >= 0; i--)  
    {  
        printf("%d ", a[i]);  
    }
```

```
    return 0;  
}
```

C:\Users\SRIRAM\OneDrive\D × + ▾

Enter number of elements: 5

Enter array elements:

12 23 45 68 79

Reversed array:

79 68 45 23 12

Process exited after 16.03 seconds with return value 0

Press any key to continue . . . |

6.Insert an element into an array

```
int main()
```

```
{  
    int a[100], n, i, pos, value;
```

```
    printf("Enter number of elements: ");  
    scanf("%d", &n);
```

```
    printf("Enter array elements:\n");  
    for(i = 0; i < n; i++)  
    {  
        scanf("%d", &a[i]);  
    }
```

```
    printf("Enter position to insert: ");  
    scanf("%d", &pos);
```

```
    printf("Enter element to insert: ");  
    scanf("%d", &value);
```

```
    for(i = n; i >= pos; i--)  
    {  
        a[i] = a[i - 1];  
    }
```

```
    a[pos - 1] = value;  
    n++;
```

```
    printf("Array after insertion:\n");  
    for(i = 0; i < n; i++)  
    {  
        printf("%d ", a[i]);  
    }
```

```
    return 0;  
}
```

C:\Users\SRIRAM\OneDrive\D × + ▾

Enter number of elements: 5

Enter array elements:

21 23 45 67 79

Enter position to insert: 3

Enter element to insert: 21

Array after insertion:

21 23 21 45 67 79

Process exited after 60.56 seconds with return value 0

Press any key to continue . . . |

7.Delete an element from an array

```
#include <stdio.h>

int main()
{
    int a[100], n, i, pos;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter position to delete: ");
    scanf("%d", &pos);

    for(i = pos - 1; i < n - 1; i++)
    {
        a[i] = a[i + 1];
    }

    n--;

    printf("Array after deletion:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

```
C:\Users\SRIRAM\OneDrive\D × + v
Enter number of elements: 5
Enter array elements:
21 25 34 65 78
Enter position to delete: 3
Array after deletion:
21 25 65 78
-----
Process exited after 32.68 seconds with return value 0
Press any key to continue . . . |
```

8.Merge two arrays

```
#include <stdio.h>

int main()
{
    int a[100], b[100], c[200];
    int n1, n2, i;

    printf("Enter number of elements in first array: ");
    scanf("%d", &n1);

    printf("Enter first array elements:\n");
    for(i = 0; i < n1; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter number of elements in second array: ");
    scanf("%d", &n2);

    printf("Enter second array elements:\n");
    for(i = 0; i < n2; i++)
    {
        scanf("%d", &b[i]);
    }

    for(i = 0; i < n1; i++)
    {
        c[i] = a[i];
    }

    for(i = 0; i < n2; i++)
    {
        c[n1 + i] = b[i];
    }

    printf("Merged array:\n");
    for(i = 0; i < n1 + n2; i++)
    {
        printf("%d ", c[i]);
    }

    return 0;
}
```

```
C:\Users\SRIRAM\OneDrive\D × + v
Enter number of elements in first array: 3
Enter first array elements:
21 34 57
Enter number of elements in second array: 3
Enter second array elements:
12 34 56
Merged array:
21 34 57 12 34 56
-----
Process exited after 12.96 seconds with return value 0
Press any key to continue . . . |
```

9. Copy one array into another

```
#include <stdio.h>

int main()
{
    int a[100], b[100], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n; i++)
    {
        b[i] = a[i];
    }

    printf("Copied array:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d ", b[i]);
    }

    return 0;
}
```

C:\Users\SRIRAM\OneDrive\D × + ▾

Enter number of elements: 5
Enter array elements:
12 23 45 67 57
Copied array:
12 23 45 67 57

Process exited after 11.42 seconds with return value 0
Press any key to continue . . . |

10. Find duplicate elements in an array

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Duplicate elements are:\n");

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[i] == a[j])
            {
                printf("%d ", a[i]);
                break;
            }
        }
    }

    return 0;
}
```

Enter number of elements: 7
Enter array elements:
12 32 43 56 21 12 467
Duplicate elements are:
12

Process exited after 24.38 seconds with return value 0
Press any key to continue . . . |